SHATTERLINE — Game Design Document v1.0

> **Working Title:** SHATTERLINE **Genre:** 2D Competitive Fighting Game **Platform:** Web (HTML5 Canvas / PixiJS + TypeScript) **Players:** 1-2 (Local Versus) **Art Direction:** Dark Cyberpunk Anime **Target:** Mechanically deep, fast-paced, stylish competitive fighter

> NOTE
>
> This is a living document. All names, values, and designs are subject to revision during development and playtesting.

1. Game Identity

Core Fantasy

"Two elite fighters clash in a shattered futuristic world — every punch, kick, and weapon strike feels devastating, every combo feels earned, and every parry feels godlike."

Design Pillars

Pillar	What It Means
Speed	60-second rounds. Aggressive pacing. No turtling. Momentum is king.
Skill Expression	Combos reward timing and knowledge, not button mashing. Perfect parries are high-risk, high-reward.
Character Identity	Two fighters on screen should move, attack, and feel completely different.
Impact	Every hit lands with weight — particles, screen flash, audio punch.
Accessibility	Easy to pick up (basic combos are simple), deep to master (advanced tech requires precision).

Inspirations

- **Devil May Cry** — Stylish combos, weapon specials, aerial juggling, spectacle.
- **Tekken** — Character-specific movesets, competitive depth, round-based structure.
- **Guilty Gear** — Combo gravity, aggressive pacing, strong character identity, visual impact.
- **Mortal Kombat** — Dedicated block button, high/low mix-ups, special meter system.

2. Technical Architecture

Technology Stack

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┌─────────────────────────────────────────────┐
│                SHATTERLINE                  │
├─────────────────────────────────────────────┤
│                                             │
│   UI Layer (Menus, HUD, Overlays)           │
│   ├── React 18 + TypeScript                 │
│   ├── CSS Modules / Tailwind                │
│   └── Menu transitions, character select, HUD │
│                                             │
│   Game Engine (Custom)                      │
│   ├── PixiJS 8 — Rendering, sprites, particles │
│   ├── Custom Game Loop — Fixed timestep @ 60fps │
│   ├── Custom Input System — Keyboard + future GP │
│   ├── Custom State Machine — Per-character FSM │
│   ├── Custom Physics — AABB hitbox/hurtbox  │
│   └── Custom Animation — Frame-based sprite sys │
│                                             │
│   Audio Layer                               │
│   ├── Howler.js or Web Audio API            │
│   └── SFX, music, hit impacts               │
│                                             │
│   Build Tooling                             │
│   ├── Vite — Fast dev server + bundling     │
│   └── TypeScript strict mode                │
│                                             │
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Why This Stack

- **PixiJS** gives hardware-accelerated WebGL rendering with a clean sprite/animation API. Fast enough for 60fps fighting game rendering.
- **Custom game loop** with fixed timestep ensures frame-consistent gameplay. Fighting games cannot have variable frame timing.
- **React** handles only menus and HUD — it never touches the game canvas.
- **TypeScript strict mode** catches bugs early in a system where off-by-one frame errors break combos.
- **Vite** provides instant hot reloading during development.

Performance Targets

- **60 FPS** — non-negotiable. The game loop runs at a fixed 60Hz tick rate.
- **Input latency** < 3 frames (50ms). Input is read at the start of each tick, before state updates.
- **No garbage collection stutter** — object pooling for particles, projectiles, and effects.

3. Combat System

3.1 Input Layout

6-button layout + directional inputs:

Button	Action	Default P1 Key	Default P2 Key
LP	Light Punch	U	Numpad 4
HP	Heavy Punch	I	Numpad 5
LK	Light Kick	J	Numpad 1
HK	Heavy Kick	K	Numpad 2
WS	Weapon Special	O	Numpad 6
ULT	Ultimate (when meter full)	P	Numpad 3
BLK	Block	L	Numpad 0
← →	Move left/right	A / D	Arrow Left / Right
↑	Jump	W	Arrow Up
↓	Crouch	S	Arrow Down

> **IMPORTANT**
>
> All controls must be **remappable** in Settings. These are defaults only.

3.2 Movement

Action	Input	Properties
Walk Forward	→	Speed varies per character
Walk Backward	←	Slightly slower than forward walk
Jump	↑	Fixed arc, height varies per character
Forward Jump	↑→	Diagonal arc forward
Backward Jump	↑←	Diagonal arc backward
Crouch	↓	Lowers hurtbox, enables low attacks
Forward Dash	→→ (double tap)	Quick burst of forward movement, brief recovery
Backward Dash	←← (double tap)	Quick evasive backstep, slightly slower

Movement Design Notes:

- Dashes have **startup frames** (can't act during) and **recovery frames** (vulnerable after).
- No air dashes in the base system — this keeps air approaches committal.
- Jump-in attacks are the primary way to open up a blocking opponent from above.

3.3 Attack Properties

Standing Attacks (High)

Attack	Startup	Active	Recovery	Properties
Stand LP	4f	3f	6f	Fast jab, low damage, chains into itself
Stand HP	10f	4f	14f	Slow, high damage, causes wall-push on hit
Stand LK	5f	3f	7f	Quick mid-range kick
Stand HK	12f	5f	16f	**Launcher** — sends opponent airborne on hit

Crouching Attacks (Low)

Attack	Startup	Active	Recovery	Properties
Crouch LP	4f	3f	6f	Low jab
Crouch HP	11f	4f	15f	Low sweep, knocks down
Crouch LK	5f	3f	8f	Quick low poke
Crouch HK	14f	5f	18f	**Slide/sweep** — knocks down, long range, very punishable

Aerial Attacks

Attack	Startup	Active	Recovery	Properties
Jump LP	4f	6f	Land	Quick air jab, good for starting air combos
Jump HP	8f	5f	Land	Downward slam, can extend air combos
Jump LK	5f	5f	Land	Diagonal kick, good cross-up tool
Jump HK	10f	6f	Land	Heavy air kick, ground bounces on hit

> NOTE
>
> Frame data above is **baseline**. Each character modifies these values. The Fast Martial Artist will have faster startup across the board; the Heavy Bruiser will have slower startup but more damage and active frames.

3.4 Defensive Mechanics

Blocking

- **Input:** Hold BLK button.
- **Standing Block:** Blocks all **high** attacks. Does NOT block **low** attacks.
- **Crouching Block:** Hold BLK + ↓. Blocks all **low** attacks. Does NOT block **high** attacks and jump-ins.
- **Block Stun:** Defender is locked in block animation for a set number of frames after blocking an attack. Heavy attacks cause more block stun.
- **Chip Damage:** Weapon Specials deal 10% of their normal damage when blocked. Normal attacks deal 0 chip.
- **Guard Break:** If a player blocks 8+ consecutive hits without leaving block, the next hit causes a **Guard Crush** — the defender staggers, fully vulnerable for 20 frames.

Perfect Parry

- **Input:** Press BLK within a **3-frame window** before an attack connects.
- **On success:**
  - Attacker is staggered for 16 frames (fully punishable).
  - Defender recovers instantly.
  - **+25% special meter** awarded to the defender.
  - Screen flash effect + distinctive sound.
- **On failure (pressed too early / too late):**
  - If within block window: normal block.
  - If outside block window: you get hit.
- **Design intent:** Perfect parries are the single highest-skill defensive option. The 3-frame window makes them genuinely difficult — you can't mash them. The reward (meter + full punish) is proportional to the risk.

Dodge

- **Input:** BLK + → (forward dodge) or BLK + ← (backward dodge).
- Forward dodge has **invincibility frames** (frames 1-8), but 12 frames of recovery.
- Backward dodge has **invincibility frames** (frames 1-6), but 8 frames of recovery.
- Dodges cannot be cancelled. If you dodge wrong, you're punished.

Throws

- **Input:** → + LP + LK (near opponent).
- Throws beat blocking. They cannot be blocked.
- **Throw tech:** If the defender inputs the same throw command within 7 frames of being thrown, the throw is escaped. Both players reset to neutral.
- **Throw range:** Short — must be very close.
- **Throw damage:** ~12% of max health.

3.5 Combo System

Combo Principles

1. **Chains:** Some light attacks chain into each other on hit (LP → LP → LK is a basic chain for most characters).
2. **Links:** After one attack's recovery, if the opponent is still in hitstun, another attack can connect. Requires precise timing.
3. **Cancels:** Certain normal attacks can be **cancelled** into Weapon Specials or the Ultimate. The cancel window is 4 frames.
4. **Launchers:** Stand HK launches the opponent into the air on hit. From there, aerial combos are possible.
5. **Air combos:** After launching, the attacker can jump and chain aerial attacks. Combo gravity applies.

Combo Gravity System

Hits in Combo	Hitstun Modifier	Effect
1-5	100%	Full hitstun — attacks chain freely
6-7	75%	Reduced hitstun — tighter link timing required
8-9	50%	Short hitstun — only fast attacks can link
10-11	25%	Minimal hitstun — almost impossible to extend
12+	0%	**Hard cap** — opponent always recovers. Combo drops.

Additional gravity rules:

- Launchers do NOT reset the combo counter.
- Ground bounces do NOT reset the combo counter.
- Gravity applies to ALL attacks, including aerials and weapon specials.
- Combo counter is displayed on screen during combos.

Damage Scaling

In addition to combo gravity (which limits combo *length*), damage also scales:

Hit #	Damage Modifier
1	100%
2	100%
3	90%
4	80%
5	70%
6	60%
7+	50% (minimum)

This means long combos are still valuable for meter gain and stage positioning, but they don't one-shot the opponent.

3.6 Special Meter System

Meter Properties

- **Range:** 0% to 100%.
- **Visual:** Bar below the health bar, fills left-to-right with a glowing effect.
- **Carries between rounds:** NO. Meter resets to 0% at the start of each round.

Meter Gain Sources

Action	Meter Gained	Notes
Landing a light attack	+3%	
Landing a heavy attack	+5%	
Landing a Weapon Special	+8%	
Completing a 5+ hit combo	+5% bonus	On top of per-hit gains
Completing a 8+ hit combo	+10% bonus	On top of per-hit gains
Successful Perfect Parry	+25%	High risk, high reward
Successful Throw	+5%	
Taking damage	+2% per hit	Small consolation meter, prevents snowballing
Whiffed attack	0%	No reward for missing
Hitting an opponent's guard	+1%	Minimal — can't farm meter by pressuring block

Meter Usage

- **100% meter → Ultimate Attack** (consumes full meter).
- No partial-meter EX moves in the base system. The meter is binary: full or not full.
- This keeps the meter system simple and makes Ultimates feel like genuine climax moments.

> **TIP**
>
> **Balance consideration:** With the meter sources above, a typical round should allow ~1 Ultimate per round if you're playing aggressively and landing combos. Perfect parries can accelerate this significantly. A player who turtles and blocks everything will build meter very slowly (only from taking damage at +2% per hit).

3.7 Ultimate Attacks

- **Activation:** Press ULT when meter is at 100%.
- **Startup:** 12 frames. The character flashes and enters a brief windup animation.
- **During startup:** The Ultimate **CAN be interrupted** — the opponent can hit you out of it if they react fast enough. This prevents raw Ultimate from being a guaranteed reversal.
- **On hit:** A **2-4 second cinematic sequence** plays. Camera zooms, effects trigger, the attack lands with full spectacle.
- **During cinematic:** Cannot be interrupted. Both players are locked.
- **On block:** The Ultimate is blockable but deals **25% chip damage** (significant but not lethal). The attacker is at heavy frame disadvantage afterward.
- **Damage:** ~30-35% of max health on hit.
- **Design intent:** Ultimates are powerful but not free. They can be blocked, they can be interrupted during startup, and even on hit they don't instantly win the round. They're a momentum-shifting tool, not an "I win" button.

4. Prototype Characters

4.1 Character A: KAEL — The Fast Martial Artist

Identity

- **Archetype:** Rushdown / pressure character.
- **Fighting Style:** Rapid-fire striking. Mix of kickboxing and capoeira-inspired movement.
- **Weapon:** Twin plasma knives — short range, fast, used in rapid multi-hit weapon specials.
- **Silhouette:** Lean, athletic build. Wears a fitted tactical suit with glowing circuit lines.
- **Color Palette:** Electric blue + black.

Gameplay Identity

Kael wants to be **in your face at all times.** His attacks are fast, his combos are long, and his pressure is relentless. His weakness is low damage per hit and low health — he needs to land many attacks to win, and a few big hits can take him out.

Stats (Relative to Roster)

Stat	Value	Rating
Health	900 HP	★★☆☆☆ (Low)
Walk Speed	5.5 px/f	★★★★★ (Fastest)
Dash Speed	9 px/f	★★★★★ (Fastest)
Jump Height	180 px	★★★★☆ (High)
Attack Speed	Fast	★★★★★
Damage per Hit	Low	★★☆☆☆
Combo Potential	Very High	★★★★★
Range	Short	★★☆☆☆

Frame Data Modifications (vs Baseline)

Attack	Startup Modifier	Damage
Stand LP	3f (−1)	25
Stand HP	8f (−2)	60
Stand LK	4f (−1)	28
Stand HK (Launcher)	10f (−2)	55
Crouch LP	3f (−1)	22
Crouch HP	9f (−2)	55
Crouch LK	4f (−1)	25
Crouch HK (Sweep)	12f (−2)	50

Signature Combos

1. Basic Combo — "Blitz Chain" `LP → LP → LK → HK (launcher)`

- 4 hits, launches opponent.
- Easy execution. The bread-and-butter for beginners.
- Damage: ~130 (before scaling on follow-up).

2. Advanced Combo — "Stormbreak" `LP → LP → LK → HK (launcher) → Jump → Jump LP → Jump LP → Jump HK (ground bounce) → Crouch HP`

- 8 hits. Requires jump-cancel timing after the launcher.
- Damage: ~280 (with scaling).

3. Signature Combo — "Phantom Edge" `LP → LK → HK (launcher) → Jump → Jump LP → Jump LK → WS (aerial plasma knife barrage) → Jump HK`

- 8 hits. Incorporates the weapon special mid-combo.
- The plasma knife barrage hits 3 times rapidly in the air.
- Damage: ~320 (with scaling).
- Visually spectacular — the knives leave blue energy trails.

Weapon Special — "Plasma Fangs"

- **Input:** WS button (can be used standing, crouching, or airborne).
- **Ground version:** A rapid 3-hit knife slash combo. 8f startup, +2 on block.
- **Air version:** Downward diagonal knife slash barrage (3 hits). Can extend air combos.
- **Damage:** 70 (ground), 55 (air, ×3 hits).

Ultimate Attack — "Lightspeed Requiem"

- Kael dashes forward at extreme speed (12f startup, punishable).
- On hit: A cinematic rapid-strike sequence — Kael circles the opponent at super speed, landing dozens of strikes from every angle, finishing with a plasma knife cross-slash that creates an X-shaped energy explosion.
- Damage: 300 (33% of max health).
- Duration: ~3 seconds.

4.2 Character B: VOSS — The Heavy Bruiser

Identity

- **Archetype:** Power / grappler character.
- **Fighting Style:** Brutal, deliberate strikes. Mix of pro wrestling power moves and street brawling.
- **Weapon:** A massive cybernetic gauntlet on his right arm that can charge up for devastating single hits.
- **Silhouette:** Tall, heavily muscular, imposing. Wears a torn armored vest and heavy boots.

- **Color Palette:** Crimson red + steel grey.

Gameplay Identity

Voss wants to **corner you and break you.** His attacks are slow but devastating. A single heavy combo can erase a third of your health bar. His weakness is his speed — fast characters can dash around him, and his whiffed attacks are heavily punishable.

Stats (Relative to Roster)

Stat	Value	Rating
Health	1200 HP	★★★★★ (Highest)
Walk Speed	3.0 px/f	★☆☆☆☆ (Slowest)
Dash Speed	5.5 px/f	★★☆☆☆ (Slow)
Jump Height	140 px	★★☆☆☆ (Low)
Attack Speed	Slow	★★☆☆☆
Damage per Hit	Very High	★★★★★
Combo Potential	Low-Medium	★★☆☆☆
Range	Long	★★★★☆

Frame Data Modifications (vs Baseline)

Attack	Startup Modifier	Damage
Stand LP	6f (+2)	40
Stand HP	14f (+4)	110
Stand LK	7f (+2)	45
Stand HK (Launcher)	16f (+4)	100
Crouch LP	6f (+2)	35
Crouch HP	15f (+4)	100
Crouch LK	7f (+2)	38
Crouch HK (Sweep)	18f (+4)	90

Signature Combos

1. Basic Combo — "Iron Chain" `LP → LP → HP`

- 3 hits. Simple but hits hard.
- Damage: ~180.

2. Advanced Combo — "Demolisher" `LP → LK → HK (launcher) → Jump → Jump HP (slam) → Crouch HK (sweep)`

- 5 hits. Requires timing the jump HP to connect before gravity pulls the opponent down.
- Damage: ~350 (with scaling).

3. Signature Combo — "Wrecking Protocol" `HP → WS (charged gauntlet slam) → → → HP (dash-cancel heavy punch)`

- 3 hits, but massive damage. The charged gauntlet slam has super armor (absorbs one hit without flinching).
- Damage: ~380 (with scaling).
- Visually spectacular — the gauntlet glows red, the ground cracks on impact.

Weapon Special — "Titan Fist"

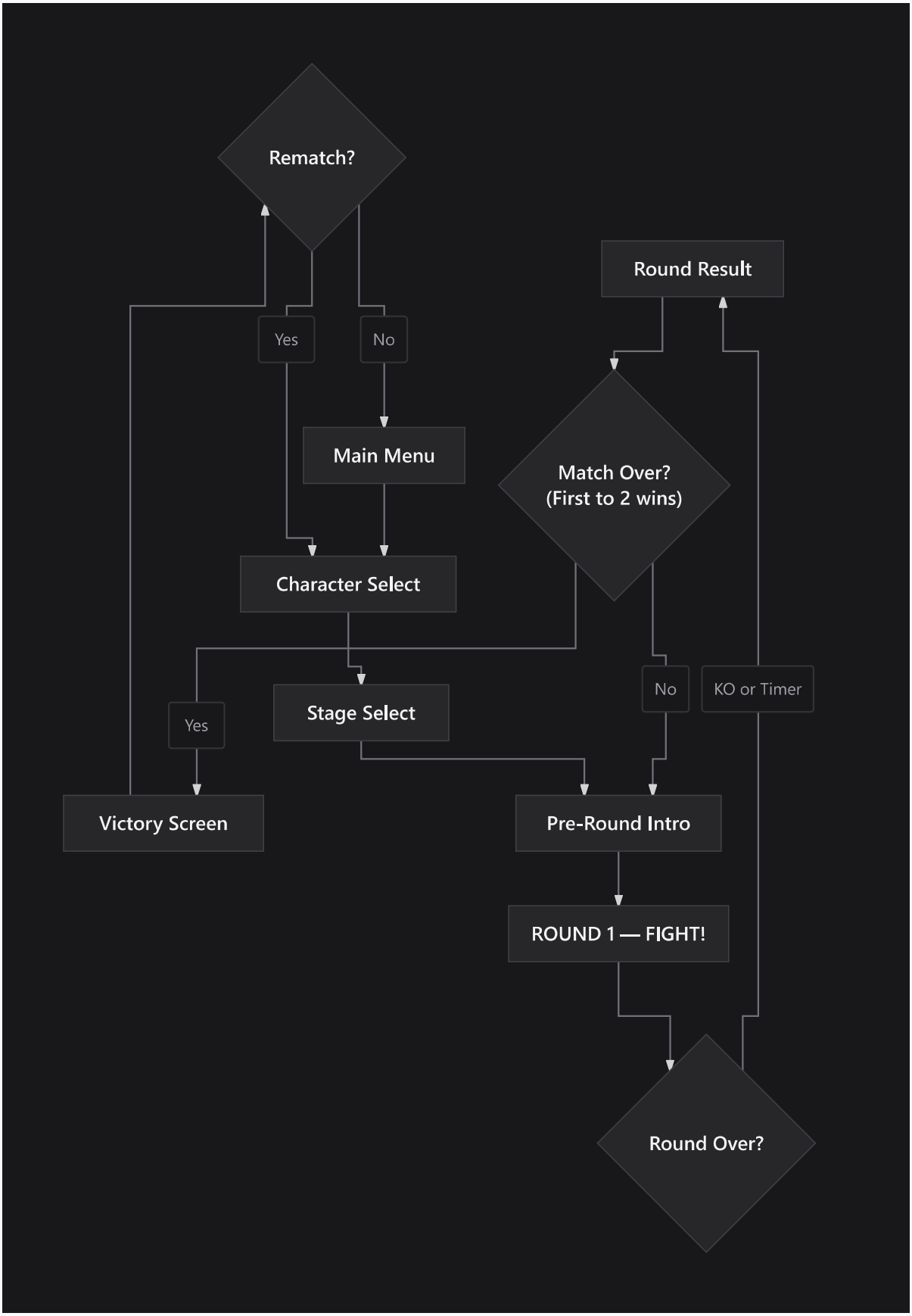
- **Input:** WS button.
- **Properties:** Voss charges his cybernetic gauntlet (hold WS to charge, release to fire).
    - **Uncharged (0-10f):** Quick forward punch. 12f startup, 80 damage.
    - **Half-charged (10-30f):** Heavier punch with super armor. 30f total, 140 damage, causes wall-bounce.
    - **Full-charged (30-60f):** Massive overhead slam. 60f total, 200 damage, causes ground bounce, **unblockable** (but extremely telegraphed and slow).
- The charge can be cancelled with a dash, creating mind games — is Voss going to release or cancel?

Ultimate Attack — "Extinction Event"

- Voss slams the ground, creating a shockwave (12f startup).
- On hit: A cinematic where Voss grabs the opponent, lifts them overhead, and pile-drives them into the ground so hard the stage cracks. He then charges his gauntlet to full power and delivers a single devastating punch that sends the opponent flying.
- Damage: 350 (29% of max health — slightly less % than Kael because Voss has more opportunities to land his due to super armor setups).
- Duration: ~3.5 seconds.

5. Match Structure

Match Flow


Rematch?
Yes
No
Main Menu
Character Select
Yes
Stage Select
Victory Screen
Round Result
Match Over?
(First to 2 wins)
No
KO or Timer
Pre-Round Intro
ROUND 1 — FIGHT!
Round Over?

Round Rules

- **Round timer:** 60 seconds.
- **Rounds per match:** Best of 3 (first to 2 wins).
- **Win condition:** Reduce opponent's HP to 0, or have more HP when timer expires.
- **Draw:** If both players have identical HP when time expires, an additional sudden-death round begins with both players at 30% HP and a 30-second timer.
- **Meter:** Resets to 0% at the start of each round.
- **Health:** Resets to 100% at the start of each round.

Stage Dimensions

- **Stage width:** 1200px (the full playable area).
- **Visible screen width:** 960px (camera follows players, scrolling when they move apart).
- **Stage height:** Grounded combat. Jump height is 140-180px above ground level.
- **Corner:** Players are pushed against the stage boundary. Corner pressure is a legitimate strategy.

6. Game Environments (4 Stages)

Stage 1: Glass Hallway

- **Setting:** The top floor of the Axiom Corporation's orbital headquarters.
- **Visuals:** Floor-to-ceiling transparent panels looking out into deep space. Neon-lit structural beams. Holographic displays flickering in the background. Stars and distant nebulae visible through the glass.
- **Color Palette:** Cold blue, white, sharp cyan accents.
- **Atmosphere:** Sterile, corporate, intimidating. The silence of space contrasts with the violence of the fight.

Stage 2: Inferno Mountain

- **Setting:** A geothermal weapons facility built into an active volcano.
- **Visuals:** Rivers of lava flowing in the background. Industrial platforms and scaffolding. Smoke and embers rising. Distant explosions from weapons testing.
- **Color Palette:** Deep red, molten orange, dark charcoal.
- **Atmosphere:** Oppressive heat, industrial danger, barely controlled chaos.

Stage 3: Neon Shore

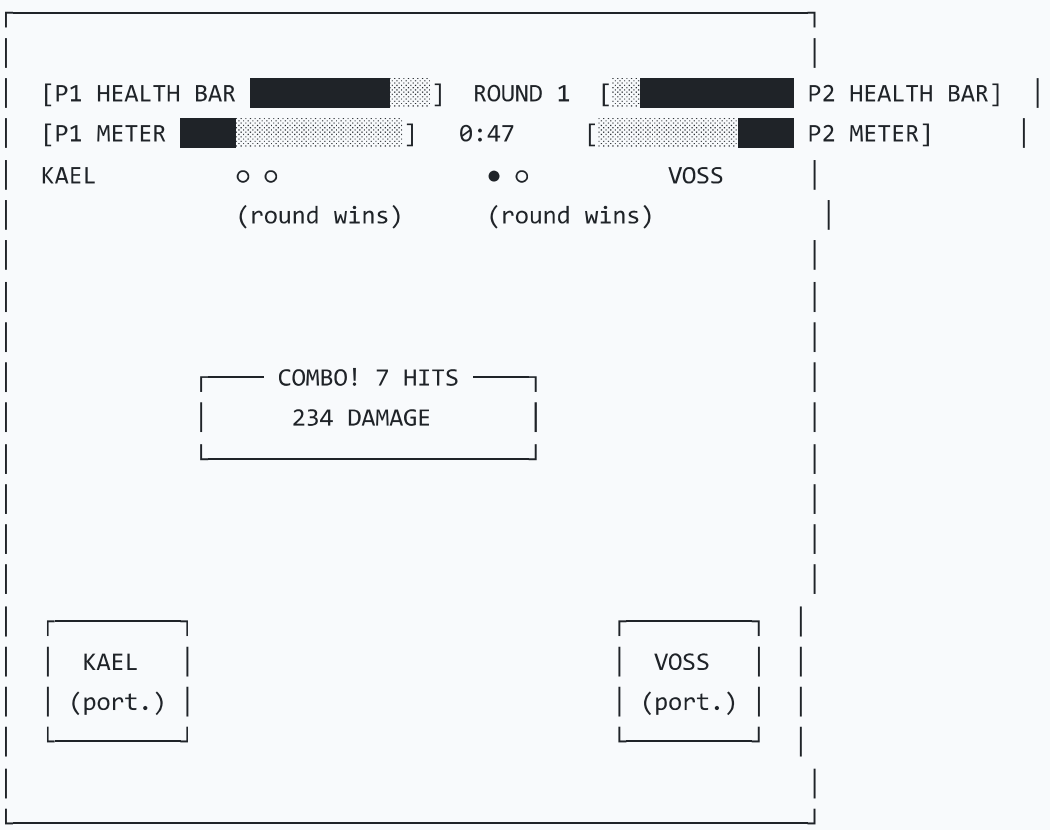
- **Setting:** A futuristic coastal resort city, abandoned after a catastrophic event.
- **Visuals:** Bioluminescent waves on a dark beach. Collapsed luxury architecture overgrown with strange vegetation. Flickering neon signs half-submerged in water. An alien moon dominates the sky.
- **Color Palette:** Dark teal, bioluminescent cyan, faded neon pink.
- **Atmosphere:** Hauntingly beautiful. Serene but wrong — something terrible happened here.

Stage 4: The Hollow *(Replaces Basketball Court)*

- **Setting:** A massive underground cavern discovered beneath a collapsed megacity.
- **Visuals:** Ancient alien ruins carved into rock walls. Strange glowing crystals providing dim light. Water dripping from the ceiling. Faint holographic symbols on the walls — a language no one can read.
- **Color Palette:** Deep purple, amber crystal glow, dark stone grey.
- **Atmosphere:** Ancient, unknowable, alien. The feeling that this place is older than human civilization and was never meant to be found.

> **TIP**
>
> Each stage will be a **parallax-scrolling background** with 3-4 depth layers moving at different speeds. This creates depth without requiring 3D rendering. PixiJS handles parallax efficiently.

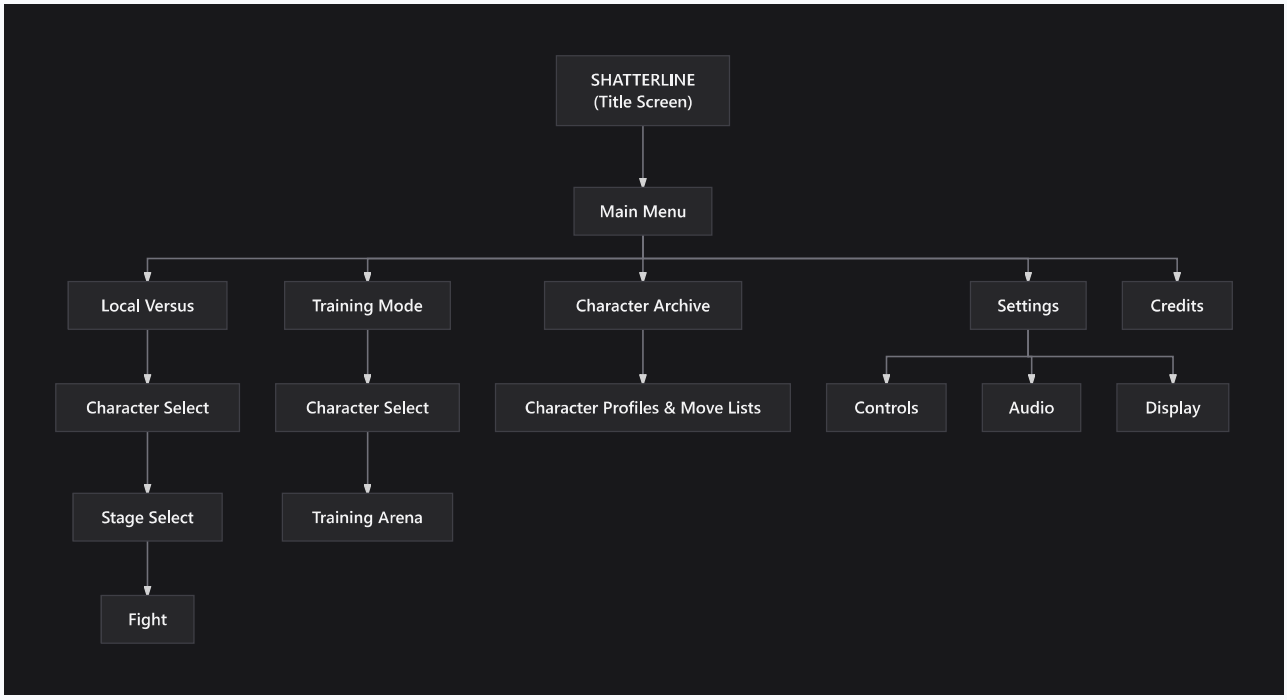
7. HUD Design



HUD Elements:

- **Health bars** — Drain with smooth animation. Flash red when below 25%.
- **Special meter** — Glows intensely at 100%. Pulses to indicate readiness.
- **Round indicators** — Filled circles for wins. First to 2.
- **Timer** — Center top. Flashes red below 15 seconds.
- **Combo counter** — Appears during combos. Shows hit count and total damage.
- **Character portraits** — Small portraits in bottom corners. React to hits and low health.

8. Menu Structure



Menu Design Principles:

- **Minimalist.** Dark background, clean typography, subtle animations.
- **Fast.** No unnecessary transitions or delays. Players should get to a fight in under 15 seconds.
- **Atmospheric.** Subtle particle effects, ambient sound, character silhouettes in the background.

> NOTE
>
> Story Mode is **deferred**. It will not be part of the initial build. When added later, it will be an arcade ladder with text/image cutscenes between fights.

9. Combat Feel ("Juice")

Confirmed Effects

Effect	Trigger	Implementation
Particle burst	Every hit connects	Spawn 8-12 particles at hit point. Color matches attacker's palette.
Screen flash	Perfect parry	Brief white flash (2 frames). Camera shake (small).
Slow-motion	Final KO hit of a match	Game speed drops to 25% for 0.5 seconds. Camera zooms slightly.
Combo counter glow	Combo exceeds 5 hits	Counter text gets larger and more vibrant.
Meter flash	Meter reaches 100%	Full-screen edge glow in character's color. Audio cue.
Hit-stop	Heavy attacks connect	3-frame pause on impact. Both characters freeze. Adds weight to big hits.

NOT Included (Deliberately)

- No dramatic screen shake on every hit — only on heavy attacks and ultimates.
- No constant slow-motion — reserved exclusively for final KO.
- No excessive particle effects — readability comes first.

10. Audio Direction

Music

- **Style:** Character-specific theme music plays on each stage. The music matches the character's personality.
- **Kael's theme:** Fast-paced electronic with sharp synths. Aggressive breakbeats. Relentless energy.
- **Voss's theme:** Heavy industrial. Deep bass hits. Mechanical percussion. Slower but punishing rhythm.
- **Implementation:** Music transitions smoothly when rounds change. Intensity increases in final round.

Sound Effects

- **Light attacks:** Sharp, quick impact sounds (snaps, cracks).
- **Heavy attacks:** Deep, weighty impacts (thuds, booms).
- **Weapon specials:** Distinct per-character weapon sounds.
- **Perfect parry:** Distinctive metallic ring — immediately recognizable.
- **Ultimate activation:** Unique audio sting per character.
- **Blocking:** Dull thud, clearly different from a clean hit.

11. Development Roadmap

Phase 1: Foundation (Current Target)

1. ✅ Finalize game design document.
2. Set up project: Vite + TypeScript + PixiJS + React.
3. Implement fixed-timestep game loop.
4. Create placeholder sprites (colored rectangles with basic animation frames).
5. Implement basic movement (walk, jump, crouch, dash).
6. Implement basic attacks (LP, HP, LK, HK with hitboxes).
7. Implement blocking (standing and crouching).

Phase 2: Combat Core

8. Implement hit detection (hitbox/hurtbox system).
9. Implement hitstun, blockstun, knockdown states.
10. Implement combo system with gravity scaling.
11. Implement damage scaling.
12. Implement launchers and aerial combos.
13. Implement perfect parry.
14. Implement throws.

Phase 3: Characters & Meter

15. Differentiate Kael and Voss (unique frame data, stats, movement).
16. Implement character-specific combo chains.

17. Implement weapon specials for both characters.
18. Implement special meter system.
19. Implement ultimate attacks.
20. Implement dodge mechanics.

Phase 4: Match Structure

21. Implement health bars and HUD.
22. Implement round system (best of 3).
23. Implement round timer (60 seconds).
24. Implement character select screen.
25. Implement stage select screen.
26. Create first parallax background (Glass Hallway).

Phase 5: Polish & Expand

27. Add hit particles and combat feel effects.
28. Add screen flash on perfect parry.
29. Add slow-motion on final KO.
30. Add hit-stop on heavy attacks.
31. Add sound effects.
32. Add music.
33. Create remaining 3 stages.
34. Balance testing and frame data tuning.

Phase 6: Full Roster (Future)

35. Design and implement remaining 4 characters.
36. Implement Training Mode.
37. Implement Character Archive.
38. Full UI polish and menu transitions.

12. Open Design Questions (For Future Sessions)

These don't need answers now, but will need to be resolved before full production:

1. **Wake-up options:** What can a knocked-down player do when getting up? (Invincible reversal? Roll? Quick rise vs delayed rise?)
2. **Wall mechanics:** What happens when a combo pushes someone into the corner? Wall-splat? Wall-bounce?
3. **Round-start spacing:** How far apart do characters start?
4. **Character select portraits:** What visual style for the UI portraits?
5. **Training mode features:** Frame data display? Hitbox visualization? Combo recording?
6. **Colorblind considerations:** Should hit effects use shapes in addition to colors?
7. **Resolution and aspect ratio:** 16:9 at what base resolution? (Recommend 960×540 scaled up.)

> IMPORTANT
>
> **This document is the contract.** Every implementation decision should reference this GDD. If something isn't in here, it doesn't get built until it's been discussed and added. This prevents scope creep — the #1 killer of indie game projects.